

Basic Shell Scripts:

=====

Create a basic script with output: Hello World!

```
[root@ip-172-31-21-46 ~]# ls
[root@ip-172-31-21-46 ~]# mkdir -p /home/vamshi/scripts
[root@ip-172-31-21-46 ~]# ls
[root@ip-172-31-21-46 ~]# cd /home
[root@ip-172-31-21-46 home]# ls
ec2-user  vamshi
[root@ip-172-31-21-46 home]# cd vamshi/
[root@ip-172-31-21-46 vamshi]# ls
scripts
[root@ip-172-31-21-46 vamshi]# cd scripts/
[root@ip-172-31-21-46 scripts]# echo $0
-bash
[root@ip-172-31-21-46 scripts]# vi helloworld.bash
[root@ip-172-31-21-46 scripts]# ls
helloworld.bash
[root@ip-172-31-21-46 scripts]# ./helloworld.bash
-bash: ./helloworld.bash: Permission denied
[root@ip-172-31-21-46 scripts]# bash ./helloworld.bash

Hello WORLD

[root@ip-172-31-21-46 scripts]# chmod 755 helloworld.bash
[root@ip-172-31-21-46 scripts]# ./helloworld.bash

Hello WORLD

[root@ip-172-31-21-46 scripts]# |
```

Basic Second script:

=====

#!/bin/bash

pwd

ls

whoami

date

cal

touch a b c

```

[root@ip-172-31-21-46 scripts]# vi userinfo.bash
[root@ip-172-31-21-46 scripts]# chmod 755 userinfo.bash
[root@ip-172-31-21-46 scripts]# ./userinfo.bash
/home/vamshi/scripts
date.bash helloworld.bash userinfo.bash
root
Thu Aug 13 12:24:45 UTC 2026
August 2026
Su Mo Tu We Th Fr Sa
      1
  2  3  4  5  6  7  8
  9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
[root@ip-172-31-21-46 scripts]# cat userinfo.bash
#!/bin/bash
# Purpose: date,calender,create files
#Date: 13/08/2026
#Modification: 13/08/2026
#Owner= vamshi

pwd
ls
whoami
date
cal
touch a b c
[root@ip-172-31-21-46 scripts]#

```

Basic Administration Tasks:

=====

top

df -h

free -m

uptime

iostat

top get stucks because that will conitnously montior.

use top | head -12

```

[root@ip-172-31-21-46 scripts]# vi admintasks.bash
[root@ip-172-31-21-46 scripts]# chmod 755 admintasks.bash
[root@ip-172-31-21-46 scripts]# ./admintasks.bash
top - 12:28:16 up 34 min, 2 users, load average: 0.00, 0.00, 0.00
tasks: 110 total, 1 running, 109 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 3.2 sy, 0.0 ni, 96.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 913.2 total, 513.5 free, 154.4 used, 245.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 627.0 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 173040 17520 10952 S   0.0   1.9   0:00.95 systemd
    2 root        20   0     0     0     0 S   0.0   0.0   0:00.00 kthreadd
    3 root        20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_release

Filesystem            Size  Used Avail Use% Mounted on
devtmpfs               420M   0  420M   0% /dev
tmpfs                  457M   0  457M   0% /dev/shm
tmpfs                  183M  436K  183M   1% /run
efivarfs               128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1         8.0G  1.7G  6.3G  22% /
tmpfs                  457M   0  457M   0% /tmp
/dev/nvme0n1p128       10M  1.3M  8.7M  13% /boot/efi
tmpfs                  92M   0   92M   0% /run/user/1000
Mem:                   total    used    free   shared  buff/cache   available
Mem:                   913      154      513        0      245      627
Swap:                   0         0         0
 12:28:16 up 34 min,  2 users,  load average: 0.00, 0.00, 0.00
Linux 6.18.39-79.141.amzn2023.x86_64 (ip-172-31-21-46.ec2.internal) 08/13/26    _x86_64_    (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.16    0.00    0.14    0.06    1.66   97.98

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1             3.93      120.98       13.25         0.00      253087     27720         0

[root@ip-172-31-21-46 scripts]# cat admintasks.bash
#!/bin/bash
# Purpose: cpu utilization, free space available,uptime
#Date: 13/08/2026
#Modification: 13/08/2026
#Owner= vamshi

top | head -10
df -h
free -m

```

Defining Variables:

=====

#!/bin/bash

#Define Variables

p=pwd

l=ls

wi=whoami

d=date

c='cal 2023'

echo Run Variables tasks

echo

\$p

\$l

\$wi

\$d

\$c

```
root@ip-172-31-21-46:/home/vamshi/scripts
root@ip-172-31-21-46 scripts]# vi variables.bash
root@ip-172-31-21-46 scripts]# chmod 755 variables.bash
root@ip-172-31-21-46 scripts]# cat variables.bash
#!/bin/bash
# Purpose: run variable tasks
#Date: 13/08/2026
#Modification: 13/08/2026
#Owner= vamshi

p=pwd
l=ls
w=whoami
d=date
c='cal 2023'

echo Run Variables tasks
echo
$P
$L
$W
$d
$c
[root@ip-172-31-21-46 scripts]# ./variables.bash
Run Variables tasks
/home/vamshi/scripts
a admintasks.bash b c date.bash helloworld.bash userinfo.bash variables.bash
root
Thu Aug 13 12:31:26 UTC 2026
2023
January February March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7 1 2 3 4 1 2 3 4
8 9 10 11 12 13 14 5 6 7 8 9 10 11 5 6 7 8 9 10 11
15 16 17 18 19 20 21 12 13 14 15 16 17 18 12 13 14 15 16 17 18
22 23 24 25 26 27 28 19 20 21 22 23 24 25 19 20 21 22 23 24 25
29 30 31 26 27 28 26 27 28 29 30 31
April May June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 1 2 3 4 5 6 1 2 3
2 3 4 5 6 7 8 7 8 9 10 11 12 13 4 5 6 7 8 9 10
9 10 11 12 13 14 15 14 15 16 17 18 19 20 11 12 13 14 15 16 17
16 17 18 19 20 21 22 21 22 23 24 25 26 27 18 19 20 21 22 23 24
23 24 25 26 27 28 29 28 29 30 31 25 26 27 28 29 30
```

Input/Output script:

=====

read --> To take command from user

echo --> to print the output

Ex1:

```
#!/bin/bash
```

```
echo Hello, my name is karthik
```

```
echo
```

```
echo what is your name?
```

```
read name
```

```
echo
```

```
echo Hello $name
```

```
echo
```

```
root@ip-172-31-21-46 scripts]# vi input.bash
root@ip-172-31-21-46 scripts]# chmod 755 input.bash
root@ip-172-31-21-46 scripts]# cat input.bash
#!/bin/bash
# Purpose: user input
#Date: 13/08/2026
#Modification: 13/08/2026
#Owner= vamshi

echo Hello, my name is sabair
echo
echo what is your name?
read name
echo
echo Hello $name
echo
[root@ip-172-31-21-46 scripts]# ./input.bash
Hello, my name is sabair

what is your name?
vamshi

Hello vamshi
[root@ip-172-31-21-46 scripts]#
```

Input/Output script:

=====

```
#!/bin/bash
```

```
#we need to use the ticks if you want to use this in echo or else not required
```

```
a=`hostname`
```

```
echo Hello, my name is $a
```

```
echo
```

```
echo what is your name?
```

```
read b
```

```
echo
```

```
echo Hello $b
```

```
echo
```

```
If i will change the a=`hostname` as the a= hostname it will does not display the ip i will
```

```
display Hello , my name is hostname
```

```
[root@ip-172-31-21-46 scripts]# vi hostname.bash
[root@ip-172-31-21-46 scripts]# chmod 755 hostname.bash
[root@ip-172-31-21-46 scripts]# cat hostname.bash
#!/bin/bash
# Purpose: display host name
#Date: 13/08/2026
#Modification: 13/08/2026
#Owner= vamshi
```

```
a=`hostname`
```

```
echo Hello, my name is $a
```

```
echo
```

```
echo what is your name?
```

```
read b
```

```
echo
```

```
echo Hello $b
```

```
echo
```

```
[root@ip-172-31-21-46 scripts]# ./hostname.bash
```

```
Hello, my name is ip-172-31-21-46.ec2.internal
```

```
what is your name?
```

```
vamshi
```

```
Hello vamshi
```

```
[root@ip-172-31-21-46 scripts]# |
```